

```
## Q_0 through Q_8
```

```
Q_0(t) = 1
Q_1(t) = t - \frac{1}{4}t
Q_2(t) = t^2 - \frac{1}{2}t + \frac{1}{4}
Q_3(t) = t^3 - \frac{1}{2}t^2 + \frac{1}{4}t - \frac{1}{8}
Q_4(t) = t^4 - \frac{1}{2}t^3 + \frac{1}{4}t^2 - \frac{1}{8}t + \frac{1}{16}
Q_5(t) = t^5 - \frac{1}{2}t^4 + \frac{1}{4}t^3 - \frac{1}{8}t^2 + \frac{1}{16}t - \frac{1}{32}
Q_6(t) = t^6 - \frac{1}{2}t^5 + \frac{1}{4}t^4 - \frac{1}{8}t^3 + \frac{1}{16}t^2 - \frac{1}{32}t + \frac{1}{64}
Q_7(t) = t^7 - \frac{1}{2}t^6 + \frac{1}{4}t^5 - \frac{1}{8}t^4 + \frac{1}{16}t^3 - \frac{1}{32}t^2 + \frac{1}{64}t - \frac{1}{128}
Q_8(t) = t^8 - \frac{1}{2}t^7 + \frac{1}{4}t^6 - \frac{1}{8}t^5 + \frac{1}{16}t^4 - \frac{1}{32}t^3 + \frac{1}{64}t^2 - \frac{1}{128}t + \frac{1}{256}
```

```
## recurrence beta,gamma,delta n=0..6
```

```
n=0: beta=1/4, gamma=0, delta=0
n=1: beta=37/108, gamma=7/144, delta=0
n=2: beta=55433/124200, gamma=355/5832, delta=23/3888
n=3: beta=17016943/40861800, gamma=1330479/21160000, delta=329/184000
n=4: beta=26633003483/59267944512, gamma=31150452736/483309595125, delta=714864/165528125
n=5: beta=1035981392580987/2411362373056448, gamma=3811477943553125/58895247500587008, delta=148436971625/60033797482491
n=6: beta=402518388816450158310763/896440355639071570400700, gamma=2780850176334226593/42650925093168016000, delta=159061031272875/40765673325633536
```

```
## values at 1 and combined second-kind numerator
```

```
n=0: Q(1)=1, P2(1)=0, P3(1)=0, R=0
n=1: Q(1)=3/4, P2(1)=1, P3(1)=1, R=2
n=2: Q(1)=4/9, P2(1)=71/108, P3(1)=115/216, R=257/216
n=3: Q(1)=3579/18400, P2(1)=50197/165600, P3(1)=2021/8640, R=533597/993600
n=4: Q(1)=31172/370125, P2(1)=611081/4568400, P3(1)=4132811/40824000, R=450896137/1918728000
n=5: Q(1)=164353085/5149165392, P2(1)=1629501983393/31777706419200, P3(1)=512081735616637/13346636696064000, R=1196472568641697/13346636696064000
n=6: Q(1)=4471493375/364303051056, P2(1)=4371727535821439/220330485278668800, P3(1)=195047569564493893/13219829116720128000, R=6535888816254319/1888547016674304000
n=7: Q(1)=21767500554751091/5000456465417001600, P2(1)=23376669819265389365867/3307801951873346558400000, P3(1)=7269637301449747792511203/1389276819786805554528000000, R=17087838625541211326175343/1389276819786805554528000000
n=8: Q(1)=3887187533860275652/2470962613391105345325, P2(1)=66976252391881601071237567/26152668300131458974919800000, P3(1)=41542213768977614279881100131/21968241372110425538932632000000, R=97802265778158159179720656411/21968241372110425538932632000000
```

```
## target initial ratios and direct sequence comparisons
```

```
target n=0: p=-612218384750, q=-215040420000, p/q=7139573/2507760
target n=1: p=-9525021973931919/18100, q=-167282265043404/905, p/q=1714499189/602214480
target n=2: p=-29561828382772029/65380, q=-964185327658080/6071, p/q=7908949578647/2778003043200
target n=3: p=-96756337931563198337/249098976, q=-5619291355848500/41187, p/q=17921159090861863/6294772017072000
target n=4: p=-14077217802009136004773/42191848160, q=-401347010628209100/3424663, p/q=2607375032785541027/915835371539088000
target n=5: p=-5998979508860456659582235/20933017687296, q=-422101321807625429375/4193312838, p/q=222225579139116749753/78056299257776112000
offset=0: initial ratio MOP=0, target=7139573/2507760, difference=-7139573/2507760
offset=1: initial ratio MOP=8/3, target=7139573/2507760, difference=-452213/2507760
offset=2: initial ratio MOP=257/96, target=7139573/2507760, difference=-852181/5015520
offset=3: initial ratio MOP=533597/193266, target=7139573/2507760, difference=-257429029/2991757680
offset=4: initial ratio MOP=450896137/161595648, target=7139573/2507760, difference=-17735053129/31268757880
offset=5: initial ratio MOP=1196472568641697/426003196320000, target=7139573/2507760, difference=-6329455188745261/164863236975840000
offset=6: initial ratio MOP=65335888816254319/23180221656000000, target=7139573/2507760, difference=-254653723972678547/8970745780872000000
```

```
## direct gauge tests
```

```
offset=0: ok=false; first=n=3: Bdiff=1328873871504055623403233743/9273485984215133441106254205
offset=1: ok=false; first=n=3: Bdiff=485400591136551908340709728062581363/3286037667442166283865348830724516800
offset=2: ok=false; first=n=3: Bdiff=20546142452540297205820050414135129094375/135385699701733930424415794098152385852776
offset=3: ok=false; first=n=3: Bdiff=17077961969225705484334063487357861904043534937265291307/111765269149989232636185163022256147368159102247991348800
offset=4: ok=false; first=n=3: Bdiff=10343199924810089023007985179473076170890342309986331306752471818215541/6703348943864900571
```

2964834198257628677235823516272034127047091512287960
offset=5: ok=false; first=n=3: Bdiff=132718077953982693335605289499877177288189685681496835139733576878375665110308584091092075/
858162847588651171265747267110405157713482341469490766603485611676409906836411750116835776
offset=6: ok=false; first=n=3: Bdiff=4908774247563218938593844639396719623681172008168597201843158558386324639491568883454661087
711477858043455823/3159074973984242810436544069466289664162604629117611293354344859443274387009374571450778558536773918712691444
0

two-step contraction gauge tests

offset=0: ok=false; first=n=3: Bdiff=954426551090661762789393986282965373813071291420551247738737125/558684595972619968046235119
4466955156081538667024299343151110032

n=2: a=58001880296730212909/312949546838894471296, b=-25771813378239925257/3422885668550408279800, c=2284417703072659913/20537
3140113024496788000

n=3: a=568278463719970596022132007934884854036/3135950409811847690063115909911194142625, b=-9283262280477880385450063736841625
465/1192041270929691437095506847694727858336, c=1648067172071105866675606142991456409/152978629769310401093923378787490075153120

n=4: a=17979572051734291293687121878563632173243694328698857185689251807941972/10014845454121042377909174330028796773778711805
1734253632067352037869985, b=-1473512423912713398981521190575007904935769559431358288037739671347359744/187371668619714650111891
918939111239930906687912597556891691192741822868125, c=2484433789453556631385353244228289905978115834188465160573146219856/23317
3632060089342361465499124227320802906100513454737465215706523157347

offset=1: ok=false; first=n=3: Bdiff=18541446350286413385018798107046815763193396812658584554733803759361992112840515685543329/1
06039749206138811143616158973270597861394269177342230462893137584069599204833993803120940

n=2: a=34048402966692285443099821902/183856261188700315229658287875, b=-487536710091611142183051295336985/60643108006678405706
479878229233516, c=12944810308943254175439500941/1650152598821181107659316414400912

n=3: a=3364324694302751441117129365173971530047523976331963991987/18548135925884416175439241848056657246616480248075109694200,
b=-1485289848335500886789233851739286940773277804908137086929/186326899810953975437067012587273659546299776673922012762500, c=8
333876334975300362490545220660078330696203741398142179/860545705134168818629491168533980107332961528574599527161440

n=4: a=9949473937082378547296477050142282821294984978389911470645432408051025404436171607066526390352677/55356306832419287603
6832864231714036991901699752710653006910960377643279608068801927142421061097472, b=-2043983894453324946940562287155861240205703
48961781616537961637596039219636831293576088974653527424/25685280653770978017731912711101917272718203537137810057767520686569926
0136043071076339482566313315005, c=10675628203352835289222629426211590494679268332700297567556310221565797476255699555701008484
3784/105315797135097714899367546000546653052561706623505076739747769155062891460742851105742383996733061875

offset=2: ok=false; first=n=3: Bdiff=2635651758243951721351455765759573490723921819983785723774236563661691296676022812412587448
2691553219/147974707825606153471834748047587692258275166833916926480801565552679445984090754665694872566883323840

n=2: a=568278463719970596022132007934884854036/3135950409811847690063115909911194142625, b=-9283262280477880385450063736841625
465/1192041270929691437095506847694727858336, c=1648067172071105866675606142991456409/152978629769310401093923378787490075153120
n=3: a=17979572051734291293687121878563632173243694328698857185689251807941972/10014845454121042377909174330028796773778711805
1734253632067352037869985, b=-1473512423912713398981521190575007904935769559431358288037739671347359744/187371668619714650111891
918939111239930906687912597556891691192741822868125, c=2484433789453556631385353244228289905978115834188465160573146219856/23317
3632060089342361465499124227320802906100513454737465215706523157347

n=4: a=531784092995715764349761882018424762711364766047127559380604743197668142127424287725960060976239898055196144109/2976188
608856366948748853566199807313310160381281449241942291497983346888455287277556958704186937860690644649984, b=-238412861454359633
0035332939831005136571315057961377414456925106862644027352198793971421651955365731600696780213125/301911268906759790106718390675
832403428637616938048094363420419711301170895403365831752394109807397198645547244839424, c=4511386292088965884477002200134101348
0354266343796153843046816525732447236872799323224737236053005581099617545/424950759928615733696290958996896481651990437405560685
7144698434786810964184165158224121846432799316795218710935431

offset=3: ok=false; first=n=3: Bdiff=1762170112585664027826957674169894405243651646409917995988813888964530830480181835978748550
65584629075584043733724980817189997377610223612863/97970601288666089581632366404237236485930363555435518887257480098486941706871
7743374959881998908501253328776201321719509419143019333405735616

offset=4: ok=false; first=n=3: Bdiff=9027739071992817455390657486840639864480515348743816919167289361402643669924748360147093494
13633219940014381675146763964726244171441942098028690871891703125/49766713251288310595702404906109071280081534268574235175915942
96870166803445166157678508609374607737487343297494006013749611404757278113866518149211016422832

offset=5: ok=false; first=n=3: Bdiff=2848516056729551335709451821138999886093138499068740457402949342824213465756470185822610759
09449477786229117288073507284768793244434467003103148975243510580155480150184445709566769327191142350860715491125/15625075699368
66499024445341325339226291779782002509740320420440184580810076524987672638222211444820369416860846086899057065560804799114571023
857258440171800792511061509938725228343603614943488550554767416

offset=6: ok=false; first=n=3: Bdiff=2674127379098426721419572576619042791933665856820887491146806599164001332260091326133977896
75046108585433576851639969669916517495152675935141777011907077526099057986945011244653241146158703726524040854899318696653413604
34161479/14600105291490655769549227069604726711514134115007498647813275461215670248784217944927260165061201585627276665186580765
8454772331013282208292695046609855854761774553030137236988040483817293876604182974983014244959430532713243840